

FTP Download & Network Configuration

Redes de Computadores

Turma 14 Grupo 7

Trabalho realizado por:

Diogo

Ramos(up202207954)

Tiago Pires(up202208910)

Sumário

O trabalho foi realizado para a cadeira de Redes de computadores, e está dividido em duas partes, a primeira consiste na implementação de um download fazendo uso do FTP, e a segunda consiste na configuração de uma rede de computadores.

Com este projeto conseguimos perceber melhor não só como funciona todo o protocolo de transferência de ficheiros, como também ter umas bases de como é configurada uma rede de computadores.

Introdução

A primeira parte do trabalho tem como objetivo desenvolver um programa de download com o protocolo de transferência de ficheiros. A segunda parte concentrou se na configuração de uma rede de computadores. O relatório estará dividido nas seguintes partes:

- **Download FTP:** Programa de download desenvolvido com FTP:
 - Funções e Arquitetura;
 - Fluxo de execução;
 - Testes.
- **Configuração da rede:**
 - Configurar uma rede IP;
 - Implementar duas bridges num switch;
 - Configurar um router em Linux;
 - Configurar um router comercial e implementar NAT;
 - DNS;
 - Ligações TCP.
- **Conclusões Finais:** Resumo de todo o trabalho feito nas duas fases do projeto e aprendizagens obtidas.
- **Anexos:** Código Download ,comandos usados nas experiências de configuração de rede e logs do wireshark de cada experiência.

Download FTP

Funções e Arquitetura

Struct URL: Armazena informações extraídas do URL: host, caminho, ficheiro, utilizador, palavra-passe e IP.

parseurl(): Extrai as partes importantes do URL fornecido pelo utilizador, e resolve o ip e o host com o uso de **gethostbyname()**, importante também para perceber se é fornecido utilizador e palavra-passe, ou se estes vão ser “anonymous”.

send_command(): Envia comandos para o servidor FTP fornecido pelo utilizador, através da socket de controlo previamente aberta.

receive_response(): Recebe e imprime na consola as respostas dadas pelo servidor, depois de serem enviados comandos.

parse_pasv_response(): Extrai o endereço ip e a porta da resposta ao comando PASV e seguidamente faz o cálculo da porta (ex: Resposta= (193,137,29,15, 211, 225) , ip = (193.137.29.15) e porta = $211 \times 256 + 225 = 54241$)

Fluxo de Execução

1. Verifica se o URL foi passado como argumento pelo utilizador, chama a função `parseurl()` para extrair as informações necessárias.
2. Cria a socket de controlo definindo o IP e a porta, e seguidamente estabelece a conexão.
3. Recebe as informações iniciais do servidor fornecido.
4. Envia o utilizador para o servidor e a palavra passe para o servidor.
5. Envia o comando para entrar em modo passivo.
6. Chama a função `parse_pasv_response()` para obter o ip e a porta a usar para a socket de dados.
7. Cria a socket de dados para a transferência, com o ip e porta do passo anterior.
8. Envia o comando RETR para pedir a o ficheiro.
9. Recebe os dados e escreve os num ficheiro local, com o mesmo nome do ficheiro do servidor.
10. Envia o comando Quit para encerrar a sessão FTP e fecha a socket.

Testes

Concluído o desenvolvimento do programa download, foram realizados vários testes, desde transferências que envolviam ficheiros de maior dimensão a ficheiros .txt pequenos. Foi testado também como é que o programa lidava com o fornecimento ou não de palavra-passe e utilizador no input. Verificou se que para todos os testes o programa tinha sido bem implementado e passado com sucesso de forma a transferir o ficheiro requisitado no input.

Na seguinte imagem percebe se como é que funciona o programa depois do input do utilizador, e seguidamente a transferência corretamente realizada:

```

dsrg@DESKTOP-U0J0H1J: /mnt/c/Users/pc/Downloads/RCOMP3032$ ./download2 ftp://demo:password@test.rebex.net/readme.txt
Host: test.rebex.net
resource: readme.txt
File: readme.txt
User: demo
Pass: password
IP: 194.188.117.16
Server: 220-Welcome to test.rebex.net!
See https://test.rebex.net/ for more information and terms of use.
220 If you don't have an account, log in as 'anonymous' or 'ftp'.
Client: USER demo
Server: 331 Anonymous login OK, send your complete email address as your password.
Client: PASS password
Server: 230 User 'demo' logged in.
Client: PASV
Server: 227 Entering Passive Mode (194,188,117,16,4,17)
Parsed PASV response:
Server IP: 194.188.117.16
Data Port: 1841
Client: RETR readme.txt
Server: 125 Data connection already open; starting 'BINARY' transfer.
File transfer completed.
Client: QUIT
Server: 226 Transfer complete.
dsrg@DESKTOP-U0J0H1J: /mnt/c/Users/pc/Downloads/RCOMP3032$ cat readme.txt
Welcome to test.rebex.net!

You are connected to an FTP or SFTP server used for testing purposes
by Rebex FTP/SSL or Rebex SFTP sample code. Only read access is allowed.

For information about Rebex FTP/SSL, Rebex SFTP and other Rebex libraries
for .NET, please visit our website at https://www.rebex.net/

For feedback and support, contact support@rebex.net

Thanks!

```

Configuração da rede

Nota:

1. Por lapso, usámos Y=2 em todas as experiências, em vez de Y=3 (número da bancada)!!!

Configuração de uma rede Ip

O objetivo desta 1ª experiência foi a configuração de dois Ip's diferentes, um para o Tux23 e outro para o Tux24, ligados no switch. Foi ainda analisado o comportamento quando as tabelas arp são eliminadas para os ip's configurados anteriormente.

Esta experiência foi muito simples, ligou se o E1 de cada computador ao switch, e depois de configurar cada um com os comandos ifconfig e usar o comando ping para ver se estavam a ser trocados pacotes entre os tux's, limpámos as tabelas arp para forçar um novo request arp entre pc's e na captura do wireshark reparámos que aparecem os ips e os mac's associados a cada tux antes do início da troca de informações. O ip e mac para o tux23 são respetivamente 172.16.20.1 e 00:50:fc:ee:0e:93, e para o tux24 são 172.16.20.254 e 00:08:54:50:3f:2c.

O log referente ao comando ping está na parte dos anexos 3.1 e os comandos usados no anexos 2.1

Questões

1. O que são os pacotes ARP e para que são usados?

Os pacotes ARP são pacotes que servem para estabelecer a ligação entre um IP e um endereço MAC.

2. Quais são os endereços MAC e IP dos pacotes ARP e porquê?

São enviados 2 endereços IP: o do tux 24 destino(172.16.50.254) e o do tux 23 origem(172.16.50.01) no mesmo pacote ARP. Como resposta o tux24 envia um pacote ARP para o tux23 que contém o seu endereço MAC, 00:08:54:50:3f:2c.

3. Que pacotes o gera comando ping ?

Enquanto tem o MAC da máquina destino, o *ping* gera pacotes ARP.

Depois de realizar a ligação do endereço IP ao respetivo MAC, gera pacotes do tipo ICMP , para transferência de informação, como observado no log.

4. Quais são os endereços MAC e IP dos pacotes ping?

São os dos computadores em comunicação no momento, tux23 e tux24.

5. Como determinar se um frame Ethernet recebido é ARP, IP ou ICMP?

No wireshark, a coluna protocol mostra qual o protocolo usado, e por norma está presente nos 2 bytes iniciais do frame.

6. Como determinar o comprimento de um frame recebido?

Também no wireshark a coluna length mostra o tamanho em bytes do frame.

7. O que é a interface de loopback e por que é que ela é importante?

É uma interface virtual que permite verificar se as ligações estão bem configuradas, desde que pelo menos um dos Ip's esteja operacional.

Implementar duas bridges num switch

O objetivo da 2ª experiência foi criar basicamente duas LAN's, uma onde estavam ligados o tux23 e o tux 24, e outra com o tux22, utilizando bridges.

Como na 1ª experiência foi necessário configurar os ip's de todos os tux's com o ifconfig. Seguidamente no terminal do switch foram criadas duas bridges, bridge20 e bridge21, e removidas as portas nas quais as interfaces estavam ligadas, que estavam ligadas na bridge default, e foram adicionadas às respetivas bridges, tux24 e tux23 à bridge20 e tux22 à bridge21.

Por fim ao tentar pingar o broadcast 172.16.20.255 no tux23, o tux24 era alcançado mas o tux22 não, o que é lógico pois o tux22 não se encontra na mesma LAN. Fazendo o mesmo para o tux22, vemos que não recebe qualquer resposta, pois o tux23 e tux24 não se encontram na mesma LAN.

O logs referentes estão na parte dos anexos 3.2 e os comandos usados no anexos 2.2.

Questões

1. Como configurar a bridge20?

A bridge20 foi configurada para servir como LAN para o tux23 e tux24. Foram eliminadas as configurações e ligações default que o switch implementou aquando da ligação entre os computadores e configuramos novas portas/ligações a cada um dos computadores.

2. Quantos domínios de broadcast existem? Como podes concluir isso a partir dos registos?

Existem dois domínios de broadcast pois existem duas bridges implementadas. Analisando os logs, reparamos que o ping do tux23 conseguiu resposta do tux24 mas não do tux22, pois estão inseridos em duas bridges diferentes.

Configurar um router em Linux

O objetivo desta 3ª experiência foi transformar o tux24 num router de modo a fazer com que o tux23 e o tux 22, que não estão na mesma LAN, consigam comunicar entre eles.

Aproveitando as configurações das experiências anteriores e as bridges já criadas, nesta experiência necessitámos apenas de conectar o E2 do tux24 ao switch configurando o com o ip 172.16.21.253. Submetemos a interface E2 aos mesmo processos de remoção da bridge default e adicionámos à bridge 21.

Seguidamente adicionámos uma gateway para que o tux23 conseguisse comunicar com o tux22, e outra para o contrário.

Por fim ao usar o comando ping, conseguimos ver que agora sim já existe comunicação entre os dois computadores.

Os comandos associados estão na secção 2.3 e os logs na 3.3.

Questões

1. Que rotas existem nos tuxes? Qual é o seu significado?

Existem as defaults e para a própria rede, e depois da experiência existse uma rota no tux22 e outra no tux23, com gateway o tux24 porque é o único que se encontra nas duas bridges e funciona como router, permitindo a troca de pacotes entre os tux's.

2. Que informações contém uma entrada da tabela de encaminhamento?

Cada entrada contém o endereço de destino e respetivo gateway.

3. Que mensagens ARP e endereços MAC associados são observados e porquê?

Embora o ping seja do tux23 para o tux22, os pacotes ARP trocados incluem apenas os endereços MAC do tux23 e do tux24, e não do destino final. O tux23 tem conhecimento apenas do endereço da gateway (tux24), que é responsável por encaminhar os pacotes para o tux22.

4. Que pacotes ICMP são observados e porquê?

Os pacotes ICMP contém sempre o endereço de origem (tux23) e o endereço de destino (tux22). Como a rede está bem configurada eles são observados e com resposta.

5. Quais são os endereços IP e MAC associados aos pacotes ICMP e porquê?

Cada pacote ICMP observado através do ping no tux23 contém como endereço de origem (ip->tux23) e o endereço de destino(ip->tux22). O endereço MAC desses pacotes é o do tux24, pois é o computador que está nas duas bridges e faz a ligação entre elas.

Configurar um router comercial e NAT

Continuando o sistema já criado das experiências anteriores, nesta 4ª experiência precisamos de conectar o ether2 do router a uma porta do switch e o ether1 ao P3.12, e o objetivo foi configurar o ip para estas duas portas do router e também conseguir o acesso à internet através do router.

Para começar adicionamos a porta onde foi ligado o ether2 do router à bridge 21, e seguidamente trocamos o cabo que ligava ao switch para o router, para assim conseguir configurar os ips e as rotas defaults em cada tux de forma a conseguir dar ping em todos os tuxs e para todos os tuxs e para o router também.

Por fim testamos o fluxo da rede com redirects ativos e desativos, com as rotas a passar diretamente pelo router em vez de passar pelo tux 24, e também a desativação do NAT.

Todos os logs referentes a esta experiência estão na secção 3.4 e os comandos 2.4

Questões

1. Como configurar uma rota estática num router comercial?

Primeiro dar reset a todas as configurações, depois adicionar à bridge correspondente, e por último atribuir um ip interno e externo, na experiência fazemos isso através das portas ether1 e ether2 do router.

2. Quais são os caminhos seguidos pelos pacotes, com e sem o ICMP redirect ativado, nas experiências realizadas, e porquê?

Na primeira experiência, sem o redirect ativo, os pacotes passam sempre pelo router antes de chegar ao destino, mesmo existindo uma rota mais otimizada como era o caso do tux24, com o redirect ativo o router informa o host que existe um melhor caminho e a partir desse momento os pacotes vão no nosso caso pelo tux24.

3. Como configurar o NAT num router comercial?

Com o comando `/ip firewall nat enable 0` para ativar ou `/ip firewall nat disable 0` para desativar, no terminal do router.

4. O que faz o NAT?

O NAT consiste em reescrever, utilizando tabelas hash, os ip's de origem de um pacote, para que assim seja possível com apenas um endereço público, servir vários privados, e assim reduzir o número de endereços públicos recorrendo a endereços privados, no nosso caso era essencial para o tux23 comunicar com o server ftp que estava numa rede externa (Feup).

5. O que acontece quando tuxY3 faz ping ao servidor FTP com o NAT desativado? Porquê?

Com o NAT desativado o tux23 não recebeu qualquer resposta dos pedidos enviados, porque o router não tem capacidade de traduzir o endereço de destino num endereço público da rede válido e com isto não lhe consegue enviar as respostas.

DNS

Com toda a rede configurada e a funcionar, o objetivo desta experiência é configurar o DNS, de forma a aceder a websites da internet através do nome do domínio e não só do ip. Para nos certificarmos que estava a funcionar tudo direito realizámos ping ao google.com.

Esta experiência foi basicamente apenas mudar no ficheiro /etc/resolv.conf em cada um dos tuxs para “nameserver ip escolhido”.

Os logs referentes à experiência encontram-se na secção 3.5 e os comandos 2.5.

Questões

1. Como configurar o serviço DNS num host?

O DNS pode ser configurado adicionando a linha "nameserver <IPescolhido>" no ficheiro /etc/resolv.conf de todos os Tuxs, como explicado no guião dado.

2. Que pacotes são trocados pelo DNS e que informação é transportada?

Antes de qualquer protocolo, inicialmente só existem pacotes do tipo DNS, uma vez que, o router precisa de identificar e traduzir o endereço de ip destino.

Ligações TCP

Para esta experiência, fizemos uso do computador pessoal para testar a nossa aplicação, uma vez que quando testámos a nossa aplicação no tux a rede da feup estava em baixo.

Os logs referentes a esta experiência encontram-se na secção 3.6

Questões

1. Quantas ligações TCP são abertas pela sua aplicação FTP?

São abertas duas ligações quando executo o download, uma de controlo (para enviar comandos entre servidor e cliente e vice versa) e uma de dados (para receber o ficheiro).

2. Em qual conexão é transportada a informação de controlo do FTP?

Na primeira descrita em cima, existe troca de comandos de controlo entre servidor e cliente.

3. Quais são as fases de uma conexão TCP?

Divide-se em 3 fases, estabelecimento da ligação (SYN + ACK), seguidamente dá-se a transferência do ficheiro e por último o fecho da ligação (FIN ACK).

4. Como funciona o mecanismo ARQ (Automatic Repeat Request) do TCP? Quais são os campos relevantes do TCP? Que informações relevantes podem ser observadas nos registos?

O mecanismo é essencial em redes onde existe perda de pacotes e onde é necessária retransmissão. Quando um pacote é enviado, o remetente aguarda um ACK. Se não houver confirmação ou se o recetor enviar duplicados, o pacote é retransmitido. O principal campo que se vê nos logs é o número de sequência e flags como ACK e SYN. Nos logs podemos ver claramente quando os ack's são duplicados e por isso têm de ser retransmitidos, como é o caso da secção 3.6.5.

5. Como funciona o mecanismo de controlo de congestionamento do TCP? Quais são os campos relevantes? Como evoluiu o débito da conexão de dados ao longo do tempo? Está de acordo com o mecanismo de controlo de congestionamento do TCP?

O mecanismo serve para ajustar a taxa de envio de dados para evitar a sobrecarga da rede e cada origem determina a sua capacidade. Utiliza algoritmos como Slow Start **1**, Congestion Avoidance **2**, e Fast Retransmit **3**, para ajustar a janela de congestionamento conforme o estado da rede, como está explicado nos slides da aula teórica. Os campos importantes são o número de sequência e o tamanho da janela. O débito começa por ser pequeno **1**, aumenta muito rápido e depois diminui quando há perdas de pacotes **3**. Está de acordo, pois adapta dinamicamente a taxa para evitar congestionamento e otimizar a comunicação.

6. O débito de uma conexão de dados TCP é perturbado pela aparição de uma segunda conexão TCP? Como?

Sim, como pode ser observado através do gráfico na secção 3.6.4 comparado com o 3.6.3 ao fazer mais de uma conexão TCP o número de pacotes por segundo é aproximadamente metade para cada ligação, assim reduz a velocidade de cada uma e aumenta bastante o tempo de transferência.

Conclusões

Com a realização deste projeto ficámos a perceber melhor a como implementar e como funciona o protocolo de transferência de ficheiros na primeira parte do projeto, e na segunda parte conseguimos perceber como funciona a configuração de uma rede de computadores, o que contribuiu para perceber também melhor o slides das aulas teóricas.

Anexos

Anexo 1 - Código aplicação Download

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <netdb.h>

#define FTP_PORT 21          // Control connection port
#define BUFFER_SIZE 1024

// Structure to store parsed URL information for FTP connection
struct URL {
    char host[256];    // A little above 253 maximum
    char resrc[512];   // 512 if the url is too long, due to a very long query
    char file[256];    // A little above 255 which is the standard
    char ip[40];       // Guarantees coverage for IPv4(15) and IPv6(39)
    char user[64];     // Usual to be limited to 32 chars
    char pass[64];     // Complex passwords can reach 64 characters
};
```

```
int parseurl(const char *urlinfo, struct URL *founded) {

    // Check if username and password are provided in the URL
    int pass = (strstr(urlinfo, "@") != NULL); // Indicates presence of password
    int user = (strstr(urlinfo, ":") != NULL); // Indicates presence of username

    if (!pass || !user) {
        // URL is for anonymous login, extract host and resource path
        int len = sscanf(urlinfo, "ftp://[%^]/%s", founded->host, founded->resrc);
        if (len != 2) {
            perror("URL is invalid for anonymous login");
            return 1;
        }

        // Set default values for anonymous user and password
        strcpy(founded->user, "anonymous");
        strcpy(founded->pass, "anonymous");
    } else {
        // URL includes username and password, extract them along with host and resource path
        int len = sscanf(urlinfo, "ftp://[%^]:[%^@%^]/%s", founded->user, founded->pass, founded->host, founded->resrc);
        if (len != 4) {
            perror("URL is invalid for login with user and password");
            return 1;
        }
    }

    // Extract file name from the last part of the URL path
    strcpy(founded->file, strrchr(urlinfo, '/') + 1);

    // Resolve the host to an IP address
    struct hostent *host;
    if ((host = gethostbyname(founded->host)) == NULL) {
        perror("gethostbyname()");
        exit(-1);
    }

    // Store the IP address
    strcpy(founded->ip, inet_ntoa(*(struct in_addr *) host->h_addr));

    return 0;
}
```

```

// Function to send a command to the FTP server with an optional argument
void send_command(int sockfd, const char *command, const char *argument) {
    char buffer[BUFFER_SIZE];
    if (argument) {
        // If there's an argument, format the command with it
        snprintf(buffer, sizeof(buffer), "%s %s\r\n", command, argument);
    } else {
        // If no argument, just send the command
        snprintf(buffer, sizeof(buffer), "%s\r\n", command);
    }
    // Send the command over the socket
    write(sockfd, buffer, strlen(buffer));
    printf("Client: %s", buffer);
}

// Function to receive and print the server's response
void receive_response(int sockfd, char *buffer) {
    ssize_t bytes = read(sockfd, buffer, BUFFER_SIZE - 1);
    if (bytes > 0) {
        // Null-terminate the buffer and print the server's response
        buffer[bytes] = '\0';
        printf("Server: %s", buffer);
    } else if (bytes < 0) {
        // Handle read error
        perror("read()");
        exit(EXIT_FAILURE);
    }
}

```

```

// Function to parse the PASV response ip
void parse_pasv_response(const char *response, char *server_ip, int *data_port) {
    int a1, a2, a3, a4, p1, p2;

    // Check for "227 " at the beginning of the response
    if (strncmp(response, "227", 3) != 0) {
        fprintf(stderr, "Invalid PASV response: %s\r\n", response);
        exit(EXIT_FAILURE);
    }

    // Extract the six numbers from the response
    if (sscanf(response, "227 Entering Passive Mode (%d,%d,%d,%d,%d,%d)",
        &a1, &a2, &a3, &a4, &p1, &p2) != 6) {
        fprintf(stderr, "Failed to parse PASV response: %s\r\n", response);
        exit(EXIT_FAILURE);
    }

    // Construct the server IP address
    sprintf(server_ip, "%d.%d.%d.%d", a1, a2, a3, a4);

    // Calculate the port
    *data_port = p1 * 256 + p2;

    printf("Parsed PASV response:\r\n");
    printf("Server IP: %s\r\n", server_ip);
    printf("Data Port: %d\r\n", *data_port);
}

```

```

int main(int argc , char *argv[]) {

    int control_sock, data_sock;
    struct sockaddr_in server_addr, data_addr;
    struct hostent *server;
    char buffer[BUFFER_SIZE];
    char server_ip[16];
    int data_port;

    if(argc != 2){
        printf("Input an URL please.\r\n");
        exit(-1);
    }

    struct URL urlinf;

    if(parseurl(argv[1], &urlinf) != 0){
        printf("Error parsing URL in main\r\n");
        exit(-1);
    }

    // Print the parsed URL information

    printf("Host: %s\r\n", urlinf.host);
    printf("resource: %s\r\n", urlinf.resrc);
    printf("File: %s\r\n", urlinf.file);
    printf("User: %s\r\n", urlinf.user);
    printf("Pass: %s\r\n", urlinf.pass);
    printf("IP: %s\r\n", urlinf.ip);

    // Step 1: Create the control socket

    memset(&server_addr, 0, sizeof(server_addr));
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = inet_addr(urlinf.ip);
    server_addr.sin_port = htons(FTP_PORT);

    if ((control_sock = socket(AF_INET, SOCK_STREAM, 0)) < 0) {
        perror("Error creating control socket");
        exit(EXIT_FAILURE);
    }

    // Connect to the server
    if (connect(control_sock, (struct sockaddr *)&server_addr, sizeof(server_addr)) < 0) {
        perror("Error connecting to server");
        exit(EXIT_FAILURE);
    }
}

```

```

// Receive initial server response
receive_response(control_sock, buffer);

// Step 2: Anonymous login
send_command(control_sock, "USER", urlinf.user);
receive_response(control_sock, buffer);
send_command(control_sock, "PASS", urlinf.pass);
receive_response(control_sock, buffer);

// Step 3: Enter passive mode
send_command(control_sock, "PASV", NULL);
receive_response(control_sock, buffer);

// Step 4: Parse the PASV response
parse_pasv_response(buffer, server_ip, &data_port);

// Step 5: Create data socket
if ((data_sock = socket(AF_INET, SOCK_STREAM, 0)) < 0) {
    perror("Error creating data socket");
    exit(EXIT_FAILURE);
}

memset(&data_addr, 0, sizeof(data_addr));
data_addr.sin_family = AF_INET;
data_addr.sin_addr.s_addr = inet_addr(server_ip);
data_addr.sin_port = htons(data_port);

if (connect(data_sock, (struct sockaddr *)&data_addr, sizeof(data_addr)) < 0) {
    perror("Error connecting to data socket");
    exit(EXIT_FAILURE);
}

// Step 6: Request file transfer
send_command(control_sock, "RETR", urlinf.resrc);
receive_response(control_sock, buffer);

```

```

// Step 7: Receive file data
FILE *file = fopen(urlinf.file, "wb");
if (!file) {
    perror("Error opening file");
    close(data_sock);
    exit(EXIT_FAILURE);
}

int bytes;
while ((bytes = read(data_sock, buffer, sizeof(buffer))) > 0) {
    fwrite(buffer, 1, bytes, file);
}

if (bytes < 0) {
    perror("Error reading data from data socket");
}

fclose(file);
close(data_sock);
printf("File transfer completed.\r\n");

// Step 8: Close control connection
send_command(control_sock, "QUIT", NULL);
receive_response(control_sock, buffer);

close(control_sock);
return 0;

```

Anexo 2 – Comandos de configuração

2.1 Configuração de uma rede ip

NO PC 3

```
systemctl restart networking
```

```
ifconfig eth1 up
```

```
ifconfig eth1 172.16.20.1/24
```

```
ifconfig {
```

```
    mac 00:50:fc:ee:0e:93
```

```
    ip 172.16.20.1
```

```
}
```

comando -> ping 172.16.20.254(responde como esperado)

```
arp -d 172.16.50.254/24
```

NO PC 4

```
GTKTERM -> /system reset-configuration
```

```
systemctl restart networking
```

```
ifconfig eth1 up
```

```
ifconfig eth1 172.16.20.254/24
```

```
ifconfig {
```

```
    mac 00:08:54:50:3f:2c
```

```
    ip 172.16.20.254
```

```
}
```

comando -> ping 172.16.20.1(responde como o esperado)

2.2 Implementar duas bridges num switch

```
EXP2

NO PC2

ifconfig eth1 up
ifconfig eth1 172.16.21.1/24
ifconfig {
    mac 00:01:02:a0:f9:60
    ip 172.16.21.1
}
ping -b 172.16.21.255(broadcast)

NO SWITCH

/interface bridge add name=bridge20
/interface bridge add name=bridge21

/interface bridge port remove [find interface =ether4]
/interface bridge port remove [find interface =ether3]
/interface bridge port remove [find interface =ether2]

/interface bridge port add bridge=bridge20 interface=ether3
/interface bridge port add bridge=bridge20 interface=ether4
/interface bridge port add bridge=bridge21 interface=ether2

NO PC3
ping 172.16.20.254(responde)
ping 172.16.21.1(não responde)
ping -b 172.16.20.255(broadcast)
```

2.3 Configuração de um router em linux

```
EXP 3

NO PC4

ifconfig eth2 up
ifconfig eth2 172.16.21.253/24
ifconfig{
    mac 00:01:02:a0:fe:ae
    ip 172.16.21.253
}

sysctl net.ipv4.ip_forward=1
sysctl net.ipv4.icmp_echo_ignore_broadcasts=0

NO SWITCH

/interface bridge port remove [find interface =ether9]
/interface bridge port add bridge=bridge21 interface=ether9

NO PC 2

route add -net 172.16.20.0/24 gw 172.16.21.253

NO PC 3

route add -net 172.16.21.0/24 gw 172.16.20.254

ping 172.16.21.1(agora responde)

ping 172.16.20.254(responde)

ping 172.16.21.254(agora responde)
```

2.4 Configuração de um router comercial e NAT

```
EXP 4

NO SWITCH

/interface bridge port remove [find interface =ether17]
/interface bridge port add bridge=bridge21 interface=ether17

NO ROUTER

/system reset-configuration
|
/ip address add address=172.16.21.254/24 interface=ether2
/ip address add address=172.16.1.31/24 interface=ether1
/ip route add dst-address=172.16.20.0/24 gateway=172.16.21.253

/ip firewall nat disable 0

/ip firewall nat enable 0

NO PC 3

route add -net 172.16.1.0/24 gw 172.16.20.254

ping 172.16.21.1
ping 172.16.20.254
ping 172.16.1.31

ping 172.16.1.10

NO PC 4

route add -net 172.16.1.0/24 gw 172.16.21.254

NO PC 2

route add -net 172.16.1.0/24 gw 172.16.21.254

sysctl net.ipv4.conf.eth1.accept_redirects=0
sysctl net.ipv4.conf.all.accept_redirects=0

route del -net 172.16.20.0 gw 172.16.21.253 netmask 255.255.255.0
route add -net 172.16.20.0/24 gw 172.16.21.254

ping 172.16.20.1

traceroute -n 172.16.20.1

route del -net 172.16.20.0 gw 172.16.21.254 netmask 255.255.255.0
route add -net 172.16.20.0/24 gw 172.16.21.253

ping 172.16.20.1

traceroute -n 172.16.20.1

sysctl net.ipv4.conf.eth1.accept_redirects=1
sysctl net.ipv4.conf.all.accept_redirects=1
```

2.5 DNS

```
EXP 5

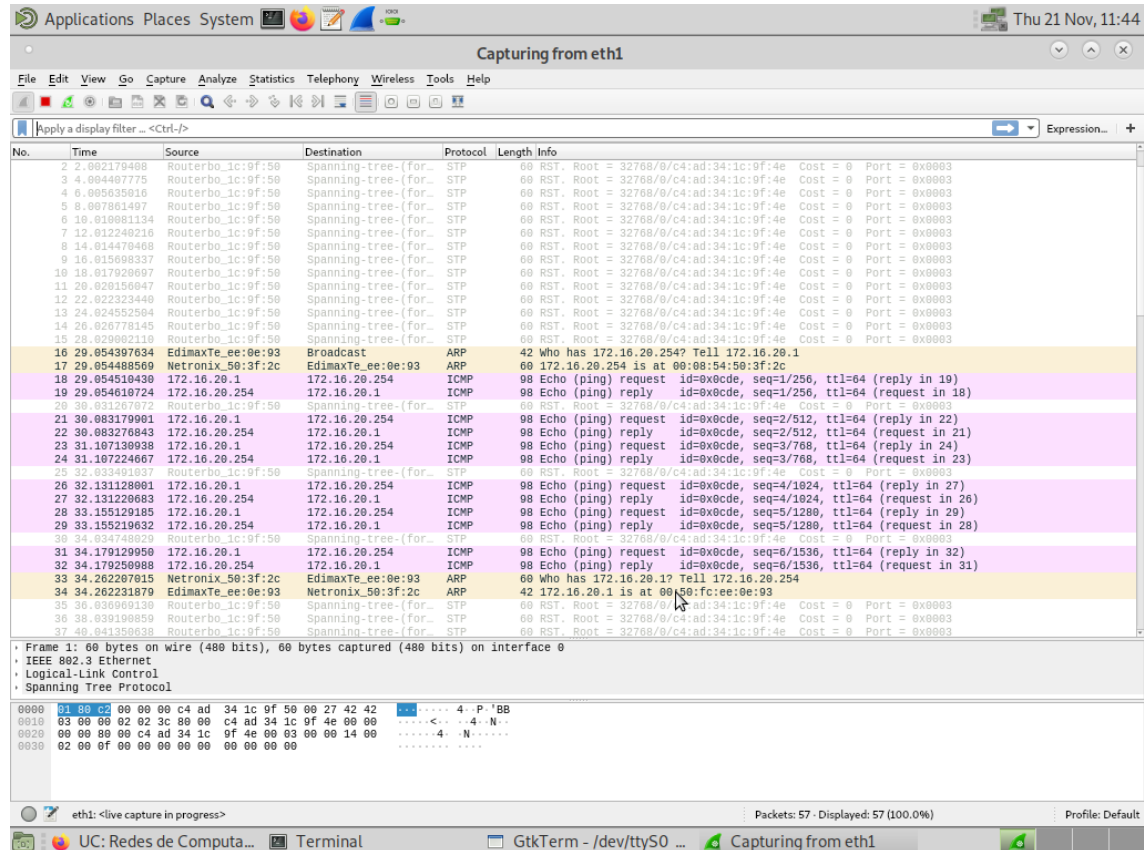
cd /etc/resolv.conf -> tudo configurado já

ping google.com(depois da configuração do dns responde)
```


Anexo 3 – Capturas Wireshark

3.1 Configuração de uma rede ip

3.1.1 Ping tux23 ao tux24



Applications Places System Thu 21 Nov, 11:44

Capturing from eth1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

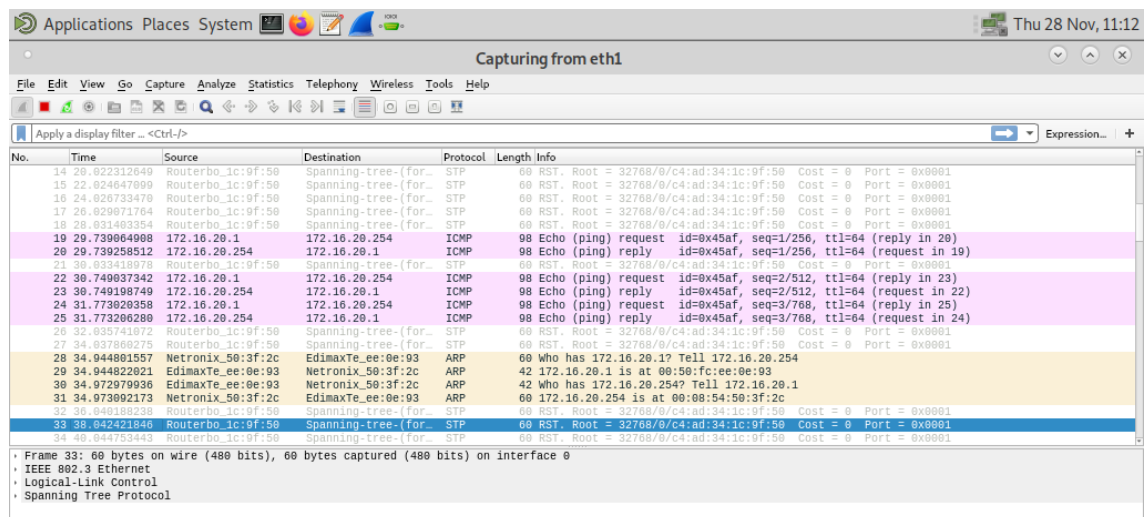
No.	Time	Source	Destination	Protocol	Length	Info
2	2.002179408	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
3	4.004407775	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
4	6.006635016	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
5	8.007861497	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
6	10.010081134	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
7	12.012240216	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
8	14.014470408	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
9	16.015698337	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
10	18.017920607	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
11	20.020156047	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
12	22.022323440	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
13	24.024552504	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
14	26.026778145	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
15	28.029002110	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
16	29.054397634	EdimaxTe_ee:0e:93	Broadcast	ARP	42	Who has 172.16.20.254? Tell 172.16.20.1
17	29.054488560	Netronix_50:3f:2c	EdimaxTe_ee:0e:93	ARP	60	172.16.20.254 is at 00:08:54:50:3f:2c
18	29.054510430	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0cde, seq=1/256, ttl=64 (reply in 19)
19	29.054610724	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0cde, seq=1/256, ttl=64 (request in 18)
20	30.031267672	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
21	30.003170901	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0cde, seq=2/512, ttl=64 (reply in 22)
22	30.003270643	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0cde, seq=2/512, ttl=64 (request in 21)
23	31.107130938	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0cde, seq=3/768, ttl=64 (reply in 24)
24	31.107224667	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0cde, seq=3/768, ttl=64 (request in 23)
25	32.033491937	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
26	32.131128001	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0cde, seq=4/1024, ttl=64 (reply in 27)
27	32.131230693	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0cde, seq=4/1024, ttl=64 (request in 26)
28	33.155129185	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0cde, seq=5/1280, ttl=64 (reply in 29)
29	33.155219632	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0cde, seq=5/1280, ttl=64 (request in 28)
30	34.034740820	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
31	34.179129950	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0cde, seq=6/1536, ttl=64 (reply in 32)
32	34.179250988	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0cde, seq=6/1536, ttl=64 (request in 31)
33	34.262207015	Netronix_50:3f:2c	EdimaxTe_ee:0e:93	ARP	60	Who has 172.16.20.1? Tell 172.16.20.254
34	34.262231879	EdimaxTe_ee:0e:93	Netronix_50:3f:2c	ARP	42	172.16.20.1 is at 00:08:54:50:3f:2c
35	35.036909130	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
36	36.039190859	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003
37	40.041350638	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0003

eth1: live capture in progress> Packets: 57 - Displayed: 57 (100.0%) Profile: Default

UC: Redes de Computa... Terminal GtkTerm - /dev/ttyS0 ... Capturing from eth1

3.2 Implementar duas bridges num switch

3.2.1 Ping tux23 ao tux24 a funcionar



Applications Places System Thu 28 Nov, 11:12

Capturing from eth1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

No.	Time	Source	Destination	Protocol	Length	Info
14	20.822312640	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
15	22.024647099	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
16	24.026733470	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
17	26.029071704	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
18	28.031404008	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
19	29.739064008	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x45af, seq=1/256, ttl=64 (reply in 20)
20	29.739258512	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x45af, seq=1/256, ttl=64 (request in 19)
21	30.833418978	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
22	30.749037342	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x45af, seq=2/512, ttl=64 (reply in 23)
23	30.749198749	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x45af, seq=2/512, ttl=64 (request in 22)
24	31.773020358	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x45af, seq=3/768, ttl=64 (reply in 25)
25	31.773206200	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x45af, seq=3/768, ttl=64 (request in 24)
26	32.835741872	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
27	34.837860275	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
28	34.944801557	Netronix_50:3f:2c	EdimaxTe_ee:0e:93	ARP	60	Who has 172.16.20.1? Tell 172.16.20.254
29	34.944822021	EdimaxTe_ee:0e:93	Netronix_50:3f:2c	ARP	42	172.16.20.1 is at 00:08:54:50:3f:2c
30	34.972979936	EdimaxTe_ee:0e:93	Netronix_50:3f:2c	ARP	42	Who has 172.16.20.254? Tell 172.16.20.1
31	34.973092173	Netronix_50:3f:2c	EdimaxTe_ee:0e:93	ARP	60	172.16.20.254 is at 00:08:54:50:3f:2c
32	36.040180238	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
33	38.042421846	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
34	40.044753443	Routerbo_1c:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001

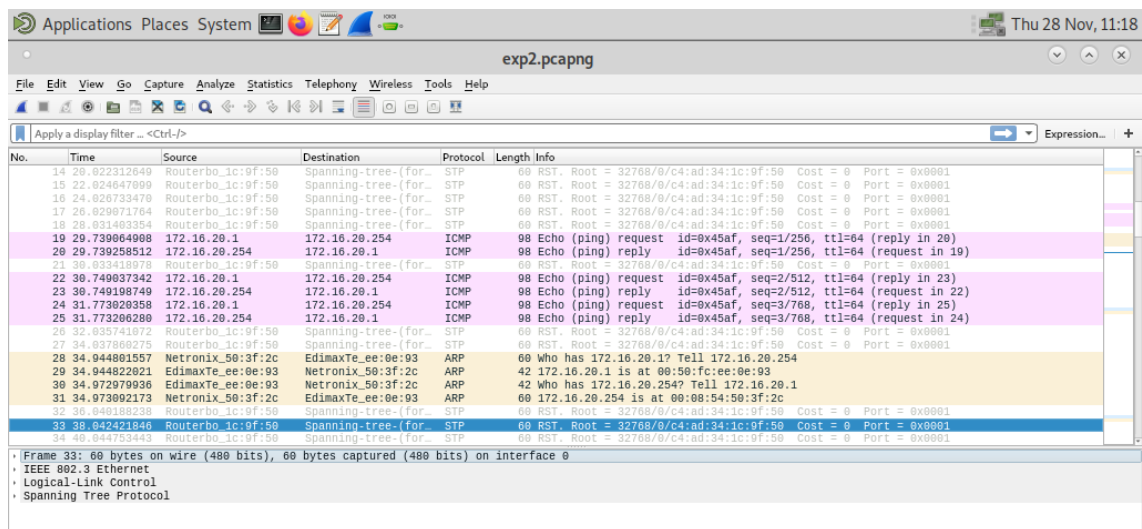
eth1: live capture in progress> Packets: 57 - Displayed: 57 (100.0%) Profile: Default

UC: Redes de Computa... Terminal GtkTerm - /dev/ttyS0 ... Capturing from eth1

3.2.2 Ping tux23 ao tux22 não funciona

```
root@tux23:~# ping 172.16.21.1
PING 172.16.21.1 (172.16.21.1) 56(84) bytes of data.
^C
--- 172.16.21.1 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 86ms
```

3.2.3 Broadcast 172.16.20.255 no tux23 com resposta do tux24



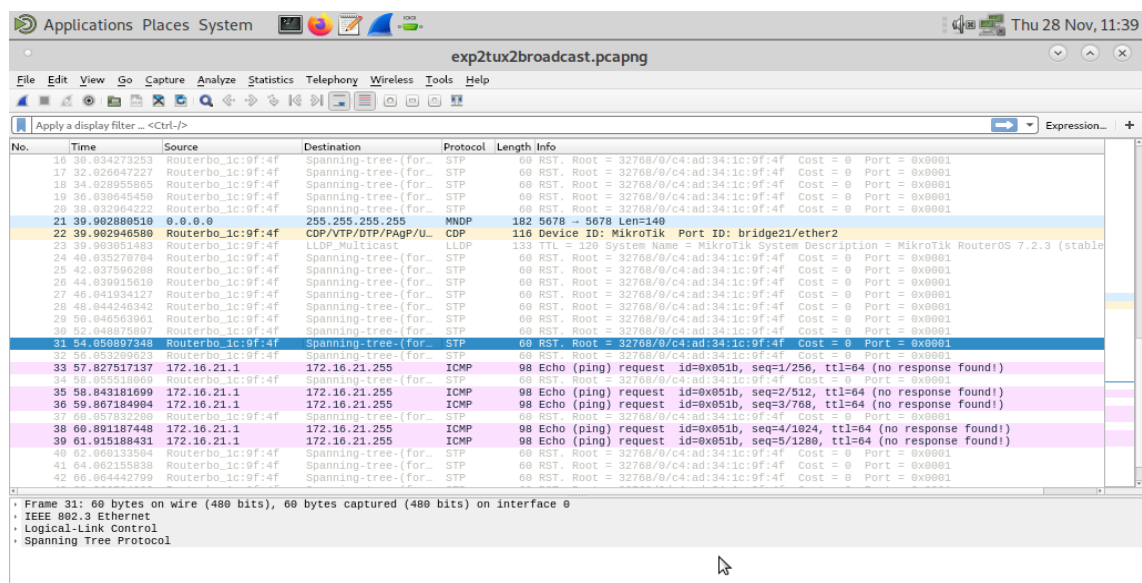
exp2.pcapng

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
14	20.822312640	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
15	22.824847000	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
16	24.826733470	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
17	26.829071764	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
18	28.831483354	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
19	29.739064908	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x45af, seq=1/256, ttl=64 (reply in 20)
20	29.739258512	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x45af, seq=1/256, ttl=64 (request in 19)
21	30.831483354	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
22	30.749073342	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x45af, seq=2/512, ttl=64 (reply in 23)
23	30.749198749	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x45af, seq=2/512, ttl=64 (request in 22)
24	31.773020358	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x45af, seq=3/768, ttl=64 (reply in 25)
25	31.773206280	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x45af, seq=3/768, ttl=64 (request in 24)
26	32.835741672	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
27	34.837869275	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
28	34.944801557	Netronix_50:3f:2c	EdimaxTe_ee:0e:93	ARP	60	Who has 172.16.20.1? Tell 172.16.20.254
29	34.944822021	EdimaxTe_ee:0e:93	Netronix_50:3f:2c	ARP	42	172.16.20.1 is at 00:50:fc:ee:0e:93
30	34.972979936	EdimaxTe_ee:0e:93	Netronix_50:3f:2c	ARP	42	Who has 172.16.20.254? Tell 172.16.20.1
31	34.973092173	Netronix_50:3f:2c	EdimaxTe_ee:0e:93	ARP	60	172.16.20.254 is at 00:08:54:90:3f:2c
32	35.841038888	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
33	35.842121645	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
34	40.844753443	Routerbo.ic:9f:50	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001

Frame 33: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
IEEE 802.3 Ethernet
Logical-Link Control
Spanning Tree Protocol

3.2.4 Broadcast 172.16.20.255 no tux22 sem resposta, e nos outros tuxs sem nada.



exp2tux2broadcast.pcapng

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
16	30.834273253	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
17	32.826647227	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
18	34.828955805	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
19	36.830845450	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
20	38.832964222	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
21	39.902800510	0.0.0.0	255.255.255.255	MNDP	192	5678 - 5678 Len=140
22	39.902940580	Routerbo.ic:9f:4f	CDP/VTP/DTP/Pag/U...	CDP	116	Device ID: Mikrotik Port ID: bridge21/ether2
23	39.903851483	Routerbo.ic:9f:4f	LLDP_Multicast	LLDP	133	TTL = 120 System Name = Mikrotik System Description = Mikrotik RouterOS 7.2.3 (stable
24	40.835270704	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
25	42.837590280	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
26	44.839919610	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
27	46.841834127	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
28	48.844240342	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
29	50.846563981	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
30	52.848878807	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
31	54.850897348	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
32	56.853209623	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
33	57.827517137	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x051b, seq=1/256, ttl=64 (no response found!)
34	59.855100809	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
35	58.843181699	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x051b, seq=2/512, ttl=64 (no response found!)
36	59.867184904	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x051b, seq=3/768, ttl=64 (no response found!)
37	60.857832280	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
38	60.891874448	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x051b, seq=4/1024, ttl=64 (no response found!)
39	61.915188431	172.16.21.1	172.16.21.255	ICMP	98	Echo (ping) request id=0x051b, seq=5/1280, ttl=64 (no response found!)
40	62.860133564	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
41	64.862155938	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001
42	66.864427999	Routerbo.ic:9f:4f	Spanning-tree-(for...	STP	60	RST, Root = 32768/0/c4:ad:34:1c:9f:4f Cost = 0 Port = 0x0001

Frame 31: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
IEEE 802.3 Ethernet
Logical-Link Control
Spanning Tree Protocol

3.3 Configuração de um router em linux

3.3.1 Ping do tux23 a todos com sucesso

5	7.189865884	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request	id=0x545f, seq=1/256, ttl=64 (reply in 6)
6	7.189243424	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x545f, seq=1/256, ttl=64 (request in 5)
7	8.008878493	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
8	8.209226671	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request	id=0x545f, seq=2/512, ttl=64 (reply in 9)
9	8.209377671	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x545f, seq=2/512, ttl=64 (request in 8)
10	8.00856423	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
11	12.013177774	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
12	14.015503454	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
13	15.541344928	172.16.20.1	172.16.21.253	ICMP	98 Echo (ping) request	id=0x5463, seq=1/256, ttl=64 (reply in 14)
14	15.541532037	172.16.21.253	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5463, seq=1/256, ttl=64 (request in 13)
15	16.017020425	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
16	16.561222572	172.16.20.1	172.16.21.253	ICMP	98 Echo (ping) request	id=0x5463, seq=2/512, ttl=64 (reply in 17)
17	16.561384677	172.16.21.253	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5463, seq=2/512, ttl=64 (request in 16)
18	17.585224917	172.16.20.1	172.16.21.253	ICMP	98 Echo (ping) request	id=0x5463, seq=3/768, ttl=64 (reply in 19)
19	17.585384647	172.16.21.253	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5463, seq=3/768, ttl=64 (request in 18)
20	18.020108177	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
21	18.609226354	172.16.20.1	172.16.21.253	ICMP	98 Echo (ping) request	id=0x5463, seq=4/1024, ttl=64 (reply in 22)
22	18.609386014	172.16.21.253	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5463, seq=4/1024, ttl=64 (request in 21)
23	20.022010241	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
24	20.561185321	EdimaxTechno_ee:0e:93	Netronix_50:3f:2c	ARP	42 Who has 172.16.20.254? Tell 172.16.20.1	
25	20.561341839	Netronix_50:3f:2c	EdimaxTechno_ee:0e:93	ARP	60 172.16.20.254 is at 00:08:54:50:3f:2c	
26	22.024341508	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
27	24.026660273	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
28	24.069807127	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request	id=0x546a, seq=1/256, ttl=64 (reply in 29)
29	24.069821883	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x546a, seq=1/256, ttl=63 (request in 28)
30	25.073226418	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request	id=0x546a, seq=2/512, ttl=64 (reply in 31)
31	25.073510399	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x546a, seq=2/512, ttl=63 (request in 30)
32	26.028981552	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
33	26.097227506	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request	id=0x546a, seq=3/768, ttl=64 (reply in 34)
34	26.097507785	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x546a, seq=3/768, ttl=63 (request in 33)
35	28.030514063	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
36	30.033251430	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001
37	30.265361553	Netronix_50:3f:2c	EdimaxTechno_ee:0e:93	ARP	60 Who has 172.16.20.1? Tell 172.16.20.254	
38	30.265377687	EdimaxTechno_ee:0e:93	Netronix_50:3f:2c	ARP	42 172.16.20.1 is at 00:50:fc:ee:0e:93	
39	32.055071123	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0001

3.3.2 Ping do tux 23 ao tux 22 refletido no eth1 e eth2 do pc

4

eth1

137	248.275989453	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0002
138	248.513383325	EdimaxTechno_ee:0e:93	Broadcast	ARP	60 Who has 172.16.20.254? Tell 172.16.20.1	
139	248.513402112	Netronix_50:3f:2c	EdimaxTechno_ee:0e:93	ARP	42 172.16.20.254 is at 00:08:54:50:3f:2c	
140	248.513515955	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request	id=0x5540, seq=1/256, ttl=64 (reply in 141)
141	248.513660276	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5540, seq=1/256, ttl=63 (request in 140)
142	249.518914103	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request	id=0x5540, seq=2/512, ttl=64 (reply in 143)
143	249.519120207	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5540, seq=2/512, ttl=63 (request in 142)
144	250.278302402	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0002
145	250.542909237	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request	id=0x5540, seq=3/768, ttl=64 (reply in 146)
146	250.543080350	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5540, seq=3/768, ttl=63 (request in 145)
147	252.280306283	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0002
148	253.650937818	Netronix_50:3f:2c	EdimaxTechno_ee:0e:93	ARP	42 Who has 172.16.20.1? Tell 172.16.20.254	
149	253.651094265	EdimaxTechno_ee:0e:93	Netronix_50:3f:2c	ARP	60 172.16.20.1 is at 00:50:fc:ee:0e:93	
150	254.282614676	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0002

eth2

137	244.275989453	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0002
138	244.508858399	3Com_a0:fe:ae	Broadcast	ARP	42 Who has 172.16.21.1? Tell 172.16.21.253	
139	244.509023575	KYE_13:4c:f3	3Com_a0:fe:ae	ARP	60 172.16.21.1 is at 00:c0:df:13:4c:f3	
140	244.509038382	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request	id=0x5540, seq=1/256, ttl=63 (reply in 141)
141	244.509175691	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5540, seq=1/256, ttl=64 (request in 140)
142	245.514261017	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request	id=0x5540, seq=2/512, ttl=63 (reply in 143)
143	245.514429337	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5540, seq=2/512, ttl=64 (request in 142)
144	246.273670059	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0002
145	246.530255243	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request	id=0x5540, seq=3/768, ttl=63 (reply in 146)
146	246.530389130	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply	id=0x5540, seq=3/768, ttl=64 (request in 145)
147	248.275674150	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0002
148	249.757886195	KYE_13:4c:f3	3Com_a0:fe:ae	ARP	60 Who has 172.16.21.253? Tell 172.16.21.1	
149	249.757907148	3Com_a0:fe:ae	KYE_13:4c:f3	ARP	42 172.16.21.253 is at 00:01:02:a0:fe:ae	
150	250.277975418	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0002
151	252.280373803	Routerboard_ic:9f:50	Spanning-tree-(for-bridges)_	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:50	Cost = 0 Port = 0x0002

3.4 Configuração de um router comercial e NAT

3.4.1 Ping do tux23 para todos

3	4.004293448	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
4	4.897766380	172.16.20.1	172.16.1.31	ICMP	98	Echo (ping) request id=0x0bdc, seq=1/256, ttl=64 (reply in 5)
5	4.898139201	172.16.1.31	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0bdc, seq=1/256, ttl=63 (request in 4)
6	5.903566550	172.16.20.1	172.16.1.31	ICMP	98	Echo (ping) request id=0x0bdc, seq=2/512, ttl=64 (reply in 7)
7	5.903890132	172.16.1.31	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0bdc, seq=2/512, ttl=63 (request in 6)
8	6.006381815	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
9	6.927564560	172.16.20.1	172.16.1.31	ICMP	98	Echo (ping) request id=0x0bdc, seq=3/768, ttl=64 (reply in 10)
10	6.927880599	172.16.1.31	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0bdc, seq=3/768, ttl=63 (request in 9)
11	7.951565363	172.16.20.1	172.16.1.31	ICMP	98	Echo (ping) request id=0x0bdc, seq=4/1024, ttl=64 (reply in 12)
12	7.951883497	172.16.1.31	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0bdc, seq=4/1024, ttl=63 (request in 11)
13	8.008498476	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
14	9.931872957	Netronix_50:3f:2c	EdimaxTechno_ee:0e:93	ARP	60	Who has 172.16.20.1? Tell 172.16.20.254
15	9.931888532	EdimaxTechno_ee:0e:93	Netronix_50:3f:2c	ARP	42	172.16.20.1 is at 00:50:fc:ee:0e:93
16	9.967519789	EdimaxTechno_ee:0e:93	Netronix_50:3f:2c	ARP	42	Who has 172.16.20.254? Tell 172.16.20.1
17	9.967635160	Netronix_50:3f:2c	EdimaxTechno_ee:0e:93	ARP	60	172.16.20.254 is at 00:08:54:50:3f:2c
18	10.010363281	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
25	16.082141568	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) request id=0x0be3, seq=1/256, ttl=64 (reply in 26)
26	16.082496300	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0be3, seq=1/256, ttl=63 (request in 25)
27	17.103567056	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) request id=0x0be3, seq=2/512, ttl=64 (reply in 28)
28	17.103873317	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0be3, seq=2/512, ttl=63 (request in 27)
29	18.019127448	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
30	18.127569118	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) request id=0x0be3, seq=3/768, ttl=64 (reply in 31)
31	18.127895005	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0be3, seq=3/768, ttl=63 (request in 30)
32	19.151583821	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) request id=0x0be3, seq=4/1024, ttl=64 (reply in 33)
33	19.151889942	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0be3, seq=4/1024, ttl=63 (request in 32)
34	20.020456979	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
35	22.023206066	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
36	24.025340918	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
37	25.466804654	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0bea, seq=1/256, ttl=64 (reply in 38)
38	25.466228227	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0bea, seq=1/256, ttl=64 (request in 37)
39	26.027446954	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
40	26.479556864	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0bea, seq=2/512, ttl=64 (reply in 41)
41	26.479705420	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0bea, seq=2/512, ttl=64 (request in 40)
42	27.503558717	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0bea, seq=3/768, ttl=64 (reply in 43)
43	27.503717330	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0bea, seq=3/768, ttl=64 (request in 42)
44	28.029598054	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
45	28.527559523	172.16.20.1	172.16.20.254	ICMP	98	Echo (ping) request id=0x0bea, seq=4/1024, ttl=64 (reply in 46)
46	28.527718453	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x0bea, seq=4/1024, ttl=64 (request in 45)
47	30.031675925	Routerboardc_1c:9f:50	Spanning-tree-(for-bri_	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001

3.4.2 Ping e traceroute do tux22 para o tux23 sem redirect

Applications Places System Thu 19 Dec, 11:58						
*eth1						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
Apply a display filter ... <Ctrl-F> Expression...						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerbo_1c:9f:4f	Spanning-tree-(for_	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
2	0.002277797	Routerbo_1c:9f:4f	Spanning-tree-(for_	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
3	3.459001824	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x2022, seq=1/256, ttl=64 (reply in 4)
4	3.459502854	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x2022, seq=1/256, ttl=63 (request in 3)
5	4.003725834	Routerbo_1c:9f:4f	Spanning-tree-(for_	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
6	4.487471783	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x2022, seq=2/512, ttl=64 (reply in 8)
7	4.487672370	172.16.20.1	172.16.21.1	ICMP	120	Redirect (Redirect for host)
8	4.487895947	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x2022, seq=2/512, ttl=63 (request in 6)
9	5.511460892	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x2022, seq=3/768, ttl=64 (reply in 11)
10	5.511665250	172.16.21.254	172.16.21.1	ICMP	120	Redirect (Redirect for host)
11	5.511933731	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x2022, seq=3/768, ttl=63 (request in 9)
12	6.005007727	Routerbo_1c:9f:4f	Spanning-tree-(for_	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
13	6.535408024	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x2022, seq=4/1024, ttl=64 (reply in 15)
14	6.535681828	172.16.21.254	172.16.21.1	ICMP	120	Redirect (Redirect for host)
15	6.535921543	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x2022, seq=4/1024, ttl=63 (request in 13)
16	8.008208587	Routerbo_1c:9f:4f	Spanning-tree-(for_	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
17	9.481435588	Routerbo_ea:74:a5	3Com_a0:f9:00	ARP	60	Who has 172.16.21.1? Tell 172.16.21.254
18	9.481457798	3Com_a0:f9:00	Routerbo_ea:74:a5	ARP	42	172.16.21.1 is at 00:01:02:a0:f9:00

```
root@tux22:~# traceroute 172.16.20.1
traceroute to 172.16.20.1 (172.16.20.1), 30 hops max, 60 byte packets
 1 172.16.21.254 (172.16.21.254) 0.449 ms 0.411 ms 0.393 ms
 2 172.16.21.253 (172.16.21.253) 0.566 ms 0.499 ms 0.525 ms
 3 172.16.20.1 (172.16.20.1) 0.639 ms 0.640 ms 0.674 ms
```


3.4.3 Ping e traceroute do tux22 para o tux23 com redirect

Applications Places System Thu 19 Dec, 12:01

exp4pc2routercomredirect.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
2	2.002225747	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
3	4.003514801	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
4	6.005538554	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
5	6.007287510	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x20f0, seq=1/256, ttl=64 (reply in 6)
6	6.007765419	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x20f0, seq=1/256, ttl=63 (request in 5)
7	7.823631886	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x20f0, seq=2/512, ttl=64 (reply in 9)
8	7.823844408	172.16.21.254	172.16.21.1	ICMP	120	Redirect (Redirect for host)
9	7.824080193	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x20f0, seq=2/512, ttl=63 (request in 7)
10	8.005551736	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
11	8.047645071	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x20f0, seq=3/768, ttl=64 (reply in 12)
12	8.047954050	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x20f0, seq=3/768, ttl=63 (request in 11)
13	9.871641138	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x20f0, seq=4/1024, ttl=64 (reply in 14)
14	9.871954866	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x20f0, seq=4/1024, ttl=63 (request in 13)
15	10.000126962	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
16	10.005648225	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x20f0, seq=5/1280, ttl=64 (reply in 17)
17	10.005967681	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x20f0, seq=5/1280, ttl=63 (request in 16)
18	11.000956688	3com_a0:fe:ae	3com_a0:f9:60	ARP	60	Who has 172.16.21.1? Tell 172.16.21.253
19	11.000958356	3com_a0:fe:ae	3com_a0:f9:60	ARP	42	172.16.21.1 is at 00:01:02:a0:f9:60
20	11.010618818	3com_a0:f9:60	Routerbo_ea:74:a5	ARP	42	Who has 172.16.21.254? Tell 172.16.21.1
21	11.010728634	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x20f0, seq=6/1536, ttl=64 (reply in 23)
22	11.0109758425	Routerbo_ea:74:a5	3com_a0:f9:60	ARP	60	172.16.21.254 is at 74:4d:28:ea:74:a5
23	11.010998991	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x20f0, seq=6/1536, ttl=63 (request in 21)
24	12.002273336	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
25	12.043637780	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x20f0, seq=7/1792, ttl=64 (reply in 26)
26	12.043953045	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x20f0, seq=7/1792, ttl=63 (request in 25)
27	13.067007188	3com_a0:f9:60	3com_a0:fe:ae	ARP	42	Who has 172.16.21.253? Tell 172.16.21.1
28	13.067408091	172.16.21.1	172.16.20.1	ICMP	98	Echo (ping) request id=0x20f0, seq=8/2048, ttl=64 (reply in 30)
29	13.067780407	3com_a0:fe:ae	3com_a0:f9:60	ARP	60	172.16.21.253 is at 00:01:02:a0:fe:ae
30	13.067925735	172.16.20.1	172.16.21.1	ICMP	98	Echo (ping) reply id=0x20f0, seq=8/2048, ttl=63 (request in 28)
31	14.003618468	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
32	16.006056998	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001
33	18.008773255	Routerbo_1c:9f:4f	Spanning-tree-(for...	STP	60	RST. Root = 32768/0/74:4d:28:ea:74:a5 Cost = 10 Port = 0x0001

```
root@tux22:~# traceroute 172.16.20.1
traceroute to 172.16.20.1 (172.16.20.1), 30 hops max, 60 byte packets
 1 172.16.21.253 (172.16.21.253) 0.216 ms 0.201 ms 0.207 ms
 2 172.16.20.1 (172.16.20.1) 0.413 ms 0.414 ms 0.364 ms
```

3.4.4 Ping FTP Server

NAT Ativo

3	3.258143603	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x10b9, seq=1/256, ttl=64 (reply in 4)
4	3.258663094	172.16.1.10	172.16.20.1	ICMP	98	Echo (ping) reply id=0x10b9, seq=1/256, ttl=62 (request in 3)
5	4.004283979	Routerboard_1c:9f:50	Spanning-tree-(for-bri...	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
6	4.285446770	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x10b9, seq=2/512, ttl=64 (reply in 7)
7	4.285858075	172.16.1.10	172.16.20.1	ICMP	98	Echo (ping) reply id=0x10b9, seq=2/512, ttl=62 (request in 6)
8	5.309448954	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x10b9, seq=3/768, ttl=64 (reply in 9)
9	5.309938622	172.16.1.10	172.16.20.1	ICMP	98	Echo (ping) reply id=0x10b9, seq=3/768, ttl=62 (request in 8)
10	6.006418522	Routerboard_1c:9f:50	Spanning-tree-(for-bri...	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
11	6.333448204	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x10b9, seq=4/1024, ttl=64 (reply in 12)
12	6.334018750	172.16.1.10	172.16.20.1	ICMP	98	Echo (ping) reply id=0x10b9, seq=4/1024, ttl=62 (request in 11)
13	7.357436628	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x10b9, seq=5/1280, ttl=64 (reply in 14)
14	7.357935935	172.16.1.10	172.16.20.1	ICMP	98	Echo (ping) reply id=0x10b9, seq=5/1280, ttl=62 (request in 13)
15	8.006544503	Routerboard_1c:9f:50	Spanning-tree-(for-bri...	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001

NAT Desativo

3	3.339313261	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x114f, seq=1/256, ttl=64 (no response found!)
4	3.356825436	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x114f, seq=2/512, ttl=64 (no response found!)
5	4.004173249	Routerboard_1c:9f:50	Spanning-tree-(for-bri...	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
6	4.30834255	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x114f, seq=3/768, ttl=64 (no response found!)
7	4.404819676	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x114f, seq=4/1024, ttl=64 (no response found!)
8	6.006264729	Routerboard_1c:9f:50	Spanning-tree-(for-bri...	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
9	6.428824025	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x114f, seq=5/1280, ttl=64 (no response found!)
10	7.452828933	172.16.20.1	172.16.1.10	ICMP	98	Echo (ping) request id=0x114f, seq=6/1536, ttl=64 (no response found!)
11	8.008399301	Routerboard_1c:9f:50	Spanning-tree-(for-bri...	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001
12	10.005113214	Routerboard_1c:9f:50	Spanning-tree-(for-bri...	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:50 Cost = 0 Port = 0x0001

3.5 DNS

3.5.1 Ping ao google.com, pacotes de dns antes dos pacotes icmp.

37	3.146248020	10.227.20.34	10.227.20.3	DNS	70 Standard query 0xf3cb A google.com
38	3.146255820	10.227.20.34	10.227.20.3	DNS	70 Standard query 0x7ad4 AAAA google.com
39	3.146762202	10.227.20.3	10.227.20.34	DNS	86 Standard query response 0xf3cb A google.com A 142.250.185.14
40	3.146780291	10.227.20.3	10.227.20.34	DNS	98 Standard query response 0x7ad4 AAAA google.com AAAA 2a00:1450:4003:803::200e
41	3.147084107	10.227.20.34	142.250.185.14	ICMP	98 Echo (ping) request id=0x1100, seq=1/256, ttl=64 (reply in 42)
42	3.164493725	142.250.185.14	10.227.20.34	ICMP	98 Echo (ping) reply id=0x1100, seq=1/256, ttl=112 (request in 41)

3.6 TCP

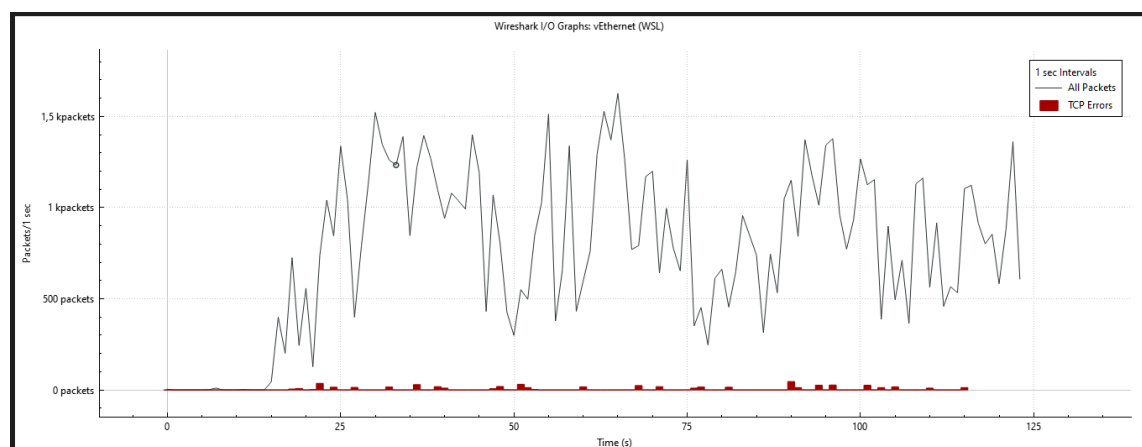
3.6.1 Início da transferência com ftp

18	15.221436	172.29.52.65	172.29.48.1	DNS	70 Standard query 0xdf02 A ftp.bit.nl
19	15.278525	172.29.48.1	172.29.52.65	DNS	96 Standard query response 0xdf02 A ftp.bit.nl A 213.136.12.213
180	15.2798081	172.29.52.65	213.136.12.213	TCP	74 45772 → 21 [SYN] Seq=0 Win=65536 Len=0 MSS=1310 SACK_PERM TSval=692158442 TSecr=0 WS=128
21	15.343179	213.136.12.213	172.29.52.65	TCP	74 21 → 45772 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3623507168 TSecr=692158442 WS=128
22	15.343647	172.29.52.65	213.136.12.213	TCP	66 45772 → 21 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=692158505 TSecr=3623507168
23	15.556306	213.136.12.213	172.29.52.65	FTP	93 Response: 220 Welcome to ftp.bit.nl
24	15.556792	172.29.52.65	213.136.12.213	TCP	66 45772 → 21 [ACK] Seq=1 Ack=28 Win=65536 Len=0 TSval=692158719 TSecr=3623507379
25	15.556991	172.29.52.65	213.136.12.213	FTP	82 Request: USER anonymous
26	15.619113	213.136.12.213	172.29.52.65	TCP	66 21 → 45772 [ACK] Seq=28 Ack=17 Win=65280 Len=0 TSval=3623507444 TSecr=692158719
27	15.620007	213.136.12.213	172.29.52.65	FTP	141 Response: 331 Anonymous login ok, send your complete email address as your password
28	15.620338	172.29.52.65	213.136.12.213	FTP	82 Request: PASS anonymous
29	15.711907	213.136.12.213	172.29.52.65	FTP	72 Response: 230-
30	15.712048	213.136.12.213	172.29.52.65	FTP	106 Response: 230-
31	15.712127	213.136.12.213	172.29.52.65	FTP	72 Response: 230-
32	15.712203	213.136.12.213	172.29.52.65	FTP	131 Response: 230-Welcome to our "PUBLIC" OPEN SOURCE SOFTWARE MIRROR SERVER.
33	15.712365	213.136.12.213	172.29.52.65	FTP	138 Response: 230-Please DO NOT report this under our responsible disclosure policy.
34	15.712441	213.136.12.213	172.29.52.65	FTP	166 Response: 230-This is a PUBLIC service, with OPEN SOURCE SOFTWARE, and NOT a security threat to our company.
35	15.712520	213.136.12.213	172.29.52.65	FTP	146 Response: 230-There is NO SENSITIVE INFORMATION on this server.
36	15.712593	213.136.12.213	172.29.52.65	FTP	122 Response: 230-
37	15.713674	172.29.52.65	213.136.12.213	TCP	66 45772 → 21 [ACK] Seq=33 Ack=528 Win=65152 Len=0 TSval=692158875 TSecr=3623507513
38	15.713674	172.29.52.65	213.136.12.213	FTP	72 Request: PASV
39	15.779441	213.136.12.213	172.29.52.65	FTP	119 Response: 227 Entering Passive Mode (213,136,12,213,169,126).
40	15.780030	172.29.52.65	213.136.12.213	TCP	74 51242 → 43390 [SYN] Seq=0 Win=65580 Len=0 MSS=1310 SACK_PERM TSval=692158942 TSecr=0 WS=128
41	15.826771	172.29.52.65	213.136.12.213	TCP	66 45772 → 21 [ACK] Seq=39 Ack=581 Win=65152 Len=0 TSval=692158909 TSecr=3623507603
42	15.894908	213.136.12.213	172.29.52.65	TCP	74 43390 → 51242 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3623507667 TSecr=692158942 WS=128
43	15.845758	172.29.52.65	213.136.12.213	TCP	66 51242 → 43390 [ACK] Seq=1 Ack=1 Win=65536 Len=0 MSS=1460 SACK_PERM TSval=3623507667 TSecr=3623507667

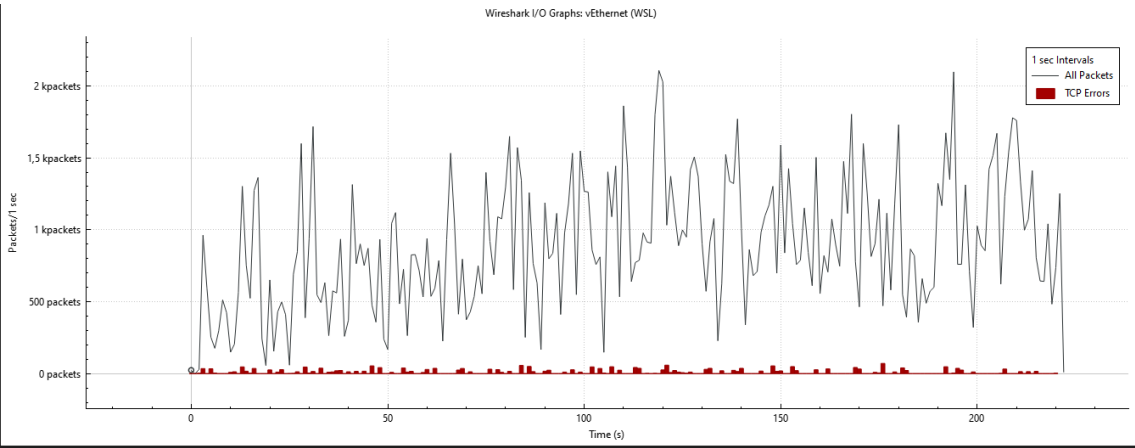
3.6.2 Fim da transferência ftp

95315	123.317552	213.136.12.213	172.29.52.65	FTP-DATA	1364 FTP Data: 1298 bytes (PASV) (USER anonymous)
95316	123.317560	213.136.12.213	172.29.52.65	FTP-DATA	772 FTP Data: 786 bytes (PASV) (USER anonymous)
95317	123.317658	172.29.52.65	213.136.12.213	TCP	66 51242 → 43390 [ACK] Seq=1 Ack=105250715 Win=1142784 Len=0 TSval=692266480 TSecr=3623615141
95318	123.318047	172.29.52.65	213.136.12.213	TCP	66 51242 → 43390 [ACK] Seq=1 Ack=105264614 Win=1138560 Len=0 TSval=692266480 TSecr=3623615141
95319	123.329477	172.29.52.65	213.136.12.213	TCP	66 51242 → 43390 [FIN, ACK] Seq=1 Ack=105264614 Win=1171712 Len=0 TSval=692266491 TSecr=3623615141
95320	123.329491	172.29.52.65	213.136.12.213	FTP	72 Request: QUIT
95321	123.389725	213.136.12.213	172.29.52.65	FTP	66 43390 → 51242 [ACK] Seq=105264614 Ack=2 Win=65280 Len=0 TSval=3623615215 TSecr=692266491
95322	123.390986	213.136.12.213	172.29.52.65	FTP	89 Response: 226 Transfer complete
95323	123.391042	213.136.12.213	172.29.52.65	FTP	88 Response: 221 Goodbye.
95324	123.391266	172.29.52.65	213.136.12.213	TCP	66 45772 → 21 [ACK] Seq=71 Ack=686 Win=65152 Len=0 TSval=692266553 TSecr=3623615216
95325	123.391577	172.29.52.65	213.136.12.213	TCP	66 45772 → 21 [FIN, ACK] Seq=71 Ack=701 Win=65152 Len=0 TSval=692266553 TSecr=3623615216
95326	123.445830	213.136.12.213	172.29.52.65	TCP	66 21 → 45772 [ACK] Seq=701 Ack=77 Win=65280 Len=0 TSval=3623615278 TSecr=692266553

3.6.3 Gráfico da transferência de 1 ficheiro, pacotes por segundo em função do tempo demorado.



3.6.4 Gráfico da transferência de 2 ficheiros ao mesmo tempo, pacotes por segundo em função do tempo demorado.



3.6.5 ACKS duplicados

538	3.128755	172.29.52.65	193.137.29.15	TCP	78	[TCP Dup ACK 525#1] 44248 → 54791 [ACK] Seq=1 Ack=434831 Win=517120 Len=0 TSval=3169966132 TSecr=1193793155
539	3.128755	172.29.52.65	193.137.29.15	TCP	78	[TCP Dup ACK 525#2] 44248 → 54791 [ACK] Seq=1 Ack=434831 Win=517120 Len=0 TSval=3169966132 TSecr=1193793155

3.6.6 Retransmissões

986	3.402542	193.137.29.15	172.29.52.65	FTP-DATA	1364	[TCP Fast Retransmission] FTP Data: 1298 bytes (P
987	3.402585	193.137.29.15	172.29.52.65	FTP-DATA	1364	[TCP Fast Retransmission] FTP Data: 1298 bytes (P